2. Semantic Density Analysis

In this notebook, we analyze semantic density as a proxy for "wooden language" in political manifestos.

The objective is to distinguish between abstract and content-rich texts using a simple quantitative measure.

```
import os
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import re
from collections import Counter
```

```
from google.colab import files
uploaded = files.upload()
```

Scegli file legislatives (1).zip

- **legislatives (1).zip**(application/x-zip-compressed) - 7384519 bytes, last modified: 28/4/2026 - 100% done

```
Saving legislatives (1).zip to legislatives (1) (2).zip
```

2.1 Loading the dataset

```
import os
import pandas as pd
import zipfile

# Assuming the uploaded zip file is 'legislatives (1) (1).zip'
zip_file_name = list(uploaded.keys())[0]

# Extract the contents of the zip file
with zipfile.ZipFile(zip_file_name, 'r') as zip_ref:
    zip_ref.extractall('/content/')

# Update the folder variable to point to the extracted directory
folder = "/content/text_files/1988/legislatives"

data = []

for filename in os.listdir(folder):
    if filename.endswith(".txt"):
        with open(os.path.join(folder, filename), "r", encoding="utf-8", errors="ignore") as f:
            text = f.read()
        data.append({"id": filename, "text": text})

df = pd.DataFrame(data)

print("Number of documents:", len(df))
df.head()
```

```
Number of documents: 3628
```

	id	text
0	EL176_L_1988_06_060_06_1_PF_01.txt	DEPARTEMENT DE L'OISE\n6ème CIRCONSCRIPTION\nC...
1	EL177_L_1988_06_095_07_1_PF_03.txt	LA FRANCE UNIE\nAVEC\nSciences Po / fonds CEVI...
2	EL175_L_1988_06_034_01_1_PF_04.txt	Sciences Po / fonds CEVIPOF\nELECTIONS LEGISLA...
3	EL177_L_1988_06_095_03_2_PF_01.txt	Sciences Po / fonds CEVIPOF\nElections législa...
4	EL176_L_1988_06_059_12_1_PF_03.txt	Sciences Po / fonds CEVIPOF\nELECTIONS LEGISLA...

Passaggi successivi: Genera codice con df New interactive sheet

2.2 Text cleaning

We normalize the text to reduce noise introduced by OCR and formatting.

```
def clean_text(text):
    text = text.lower()
    text = re.sub(r"\n+", " ", text)
    text = re.sub(r"[^a-zàâçéèêëîïôûùüÿñæœ\s]", " ", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text

df["clean_text"] = df["text"].apply(clean_text)
```

2.3 Semantic density definition

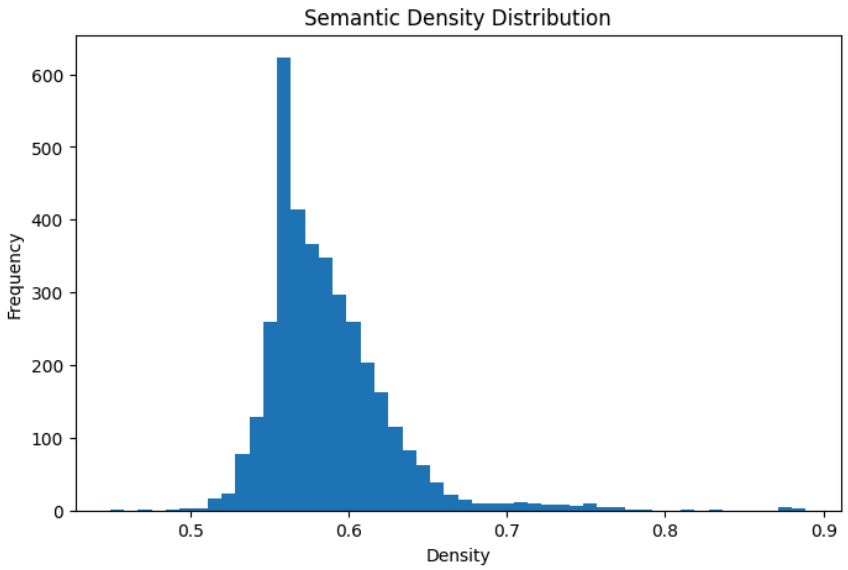
Semantic density is defined as the proportion of content words within a document.

This simplified version considers words longer than 3 characters as content words.

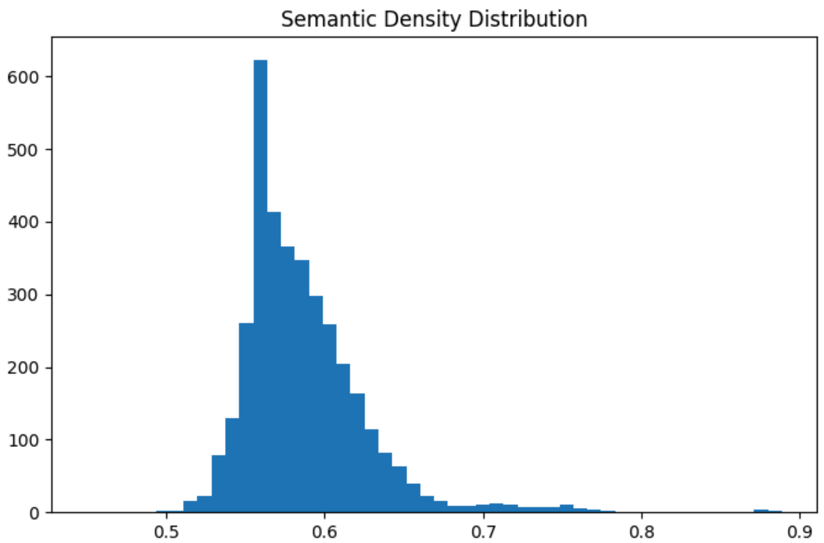
```
def semantic_density(text):
    words = text.split()
    content_words = [w for w in words if len(w) > 3]

    if len(words) == 0:
        return 0

    return len(content_words) / len(words)

df["density"] = df["clean_text"].apply(semantic_density)
```

2.4 Density distribution

```
plt.figure(figsize=(8,5))
plt.hist(df["density"], bins=50)
plt.title("Semantic Density Distribution")
plt.xlabel("Density")
plt.ylabel("Frequency")
plt.show()
```




```
plt.figure(figsize=(8,5))
plt.hist(df["density"], bins=50)
plt.title("Semantic Density Distribution")
plt.savefig("density_plot.png", dpi=300, bbox_inches="tight")
plt.show()
```



2.5 Density statistics

```
df["density"].describe()
```

	density
count	3628.000000
mean	0.586725
std	0.039760
min	0.449393
25%	0.561003
50%	0.579079
75%	0.603977
max	0.888889

dtype: float64

2.6 Quantile analysis

We analyze density distribution using quantiles to better identify extreme documents.

```
df["density"].quantile([0.1, 0.25, 0.5, 0.75, 0.9])
```

	density
0.10	0.550716
0.25	0.561003
0.50	0.579079
0.75	0.603977
0.90	0.630248

dtype: float64

2.7 Low vs High density groups

We define low-density and high-density documents using quantile thresholds.

```
low_threshold = df["density"].quantile(0.1)
high_threshold = df["density"].quantile(0.9)

low_docs = df[df["density"] <= low_threshold]
high_docs = df[df["density"] >= high_threshold]

print("Low density docs:", len(low_docs))
print("High density docs:", len(high_docs))
```

```
Low density docs: 363
High density docs: 363
```

```
low_docs[["id", "density"]].head()
```

	id	density
2	EL175_L_1988_06_034_01_1_PF_04.txt	0.544194
14	EL174_L_1988_06_004_02_1_PF_02.txt	0.546485
16	EL177_L_1988_06_095_02_1_PF_05.txt	0.548807
17	EL176_L_1988_06_071_02_1_PF_02.txt	0.550593
18	EL177_L_1988_06_083_06_2_PF_01.txt	0.520095

```
high_docs[["id", "density"]].head()
```

	id	density
20	EL174_L_1988_06_021_04_2_BV_pdfmasterocr.txt	0.666667
23	EL174_L_1988_06_005_01_1_PF_04.txt	0.656805
28	EL176_L_1988_06_059_23_1_BV_pdfmasterocr.txt	0.751381
51	EL175_L_1988_06_049_03_1_PF_02.txt	0.634868
61	EL175_L_1988_06_037_02_2_PF_02.txt	0.673016

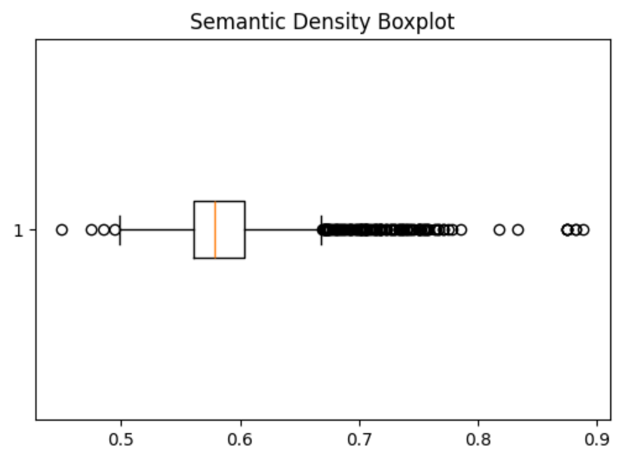
2.8 Qualitative comparison

```
print("LOW DENSITY TEXT:\n")
print(low_docs.iloc[0]["text"][:1000])

print("\n\nHIGH DENSITY TEXT:\n")
print(high_docs.iloc[0]["text"][:1000])
```

```
LOW DENSITY TEXT:

Sciences Po / fonds CEVIPOF
ELECTIONS LEGISLATIVES DE JUIN 1988, 1ère CIRCONSCRIPTION DE L'HERAULT
7 raisons de voter pour le candidat du Front National
Jean-Claude MARTINEZ
- Député de l'Hérault
- Né le 30 juillet 1945 à SETE
- Marié
- Une fille, 9 ans, Jeanne-Isabelle Occitane
- Agrégé de droit public et de Sciences po- litiques (Major de concours)
- Professeur à l'Université de Droit de Paris et à l'ENA du Maroc
- Rapporteur du budget de l'Education Na- tionale à la commission des finances
- Président de l'Association pour la sup- pression de l'impôt sur le revenu et la ré- forme fiscale (ASIREF).
Vu le Candidat
Si vous voulez un député
· qui soit présent à l'Assemblée nationale et qui remplisse effectivement le mandat pour lequel les contribuables le payent, comme les
· qui, une fois élu, ne s'entendra pas avec les socialistes contre le vœu de ses élect


HIGH DENSITY TEXT:

Sciences Po / fonds CEVIPOV
RÉPUBLIQUE FRANÇAISE QUATRIÈME CIRCONSCRIPTION DE LA CÔTE-D'OR
ÉLECTIONS LÉGISLATIVES - 12 JUIN 1988
Gilbert MATHIEU
Député sortant Vice-Président du Conseil Général de la Côte-d'Or
Union du Rassemblement et du Centre
SUPPLÉANT : Henri CONSTANT
Conseiller Général d'IS-SUR-TILLE
IMP. BORDOT - 21140 SEMURSciences Po / fonds CEVIPOV
République Française ELECTIONS LEGISLATIVES DE JUIN 1988
4· Circonscription de la Côte d'Or
MICHEL NEUGNOT
Conseiller Régional de Bourgogne Candidat de la Majorité Présidentielle pour la France Unie
SUPPLEANT : MICHEL SEREX Conseiller Municipal de Chatillon sur Seine
PRISME SA - DIJON
```

2.9 Boxplot visualization

```
plt.figure(figsize=(6,4))
plt.boxplot(df["density"], vert=False)
plt.title("Semantic Density Boxplot")
plt.show()
```

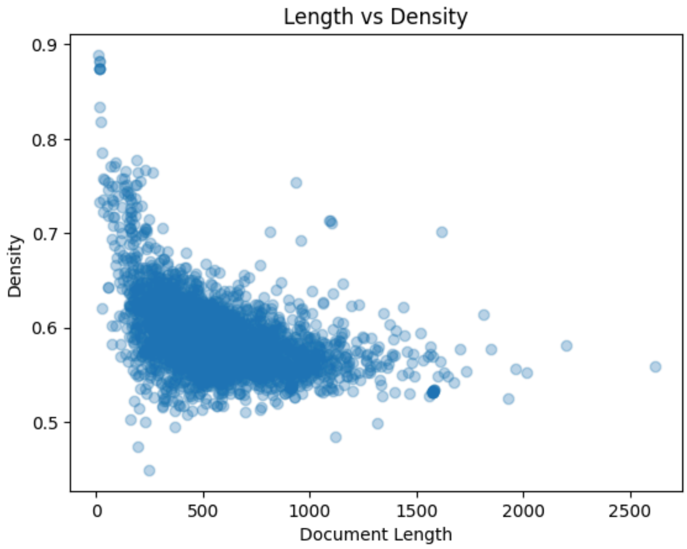


2.10 Density vs document length

```
df["length"] = df["clean_text"].apply(lambda x: len(x.split()))

df[["density", "length"]].corr()
```

	density	length
density	1.000000	-0.420071
length	-0.420071	1.000000

```
plt.scatter(df["length"], df["density"], alpha=0.3)
plt.xlabel("Document Length")
plt.ylabel("Density")
plt.title("Length vs Density")
plt.show()
```



2.11 Lexical analysis

```
low_words = " ".join(low_docs["clean_text"]).split()
high_words = " ".join(high_docs["clean_text"]).split()

print("Low density words:", Counter(low_words).most_common(10))
print("High density words:", Counter(high_words).most_common(10))
```

```
Low density words: [('de', 13995), ('la', 8666), ('et', 7368), ('l', 6188), ('le', 6119), ('à', 5356), ('les', 4881), ('des', 4439), ('
High density words: [('de', 5299), ('la', 3108), ('et', 2653), ('le', 2077), ('l', 1777), ('pour', 1769), ('du', 1679), ('à', 1480), ('
```

2.12 Interpretation

The results show that semantic density varies across documents but remains within a relatively narrow range.

Low-density texts tend to include more generic and repetitive language, while high-density texts contain more specific terms such as names, institutions, and concrete references.

The relationship between document length and density appears weak, suggesting that density captures a distinct linguistic dimension.

Overall, semantic density provides a useful proxy for distinguishing between abstract and content-rich political discourse.